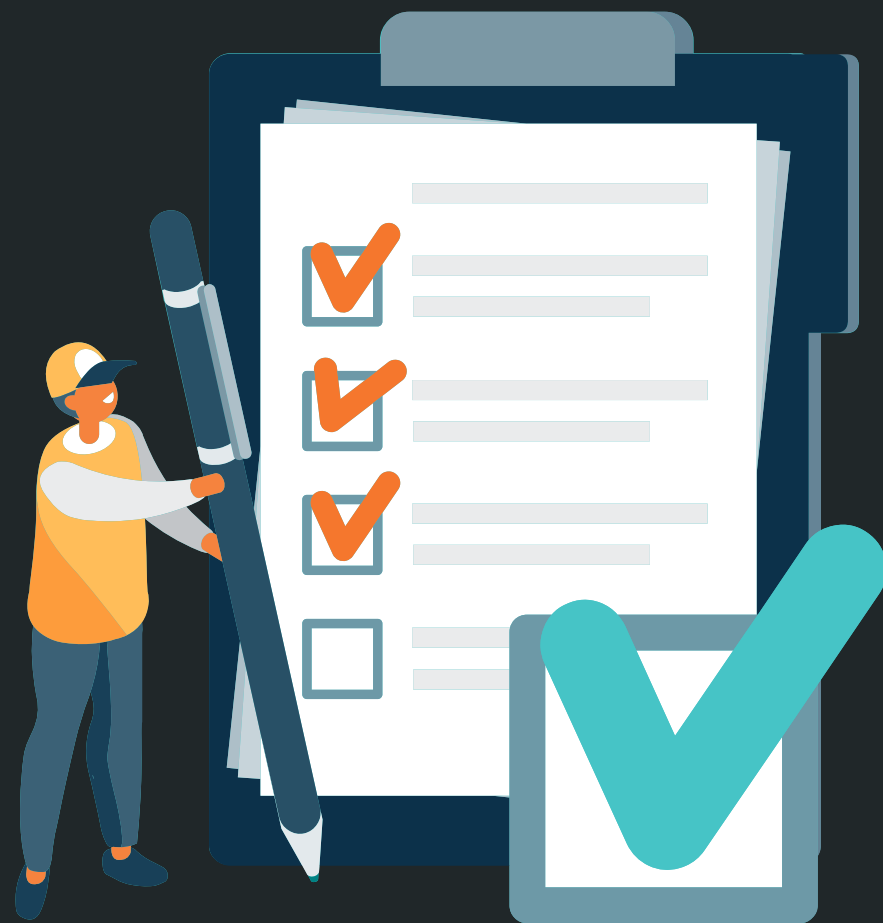


13.5. Model Optimization Checklist



Model Optimization Checklist



Model Optimization

- ✓ Improve performance step by step
- ✓ Fix data before fixing model
- ✓ Optimize systematically

Data Comes First



Better Data Handling

- ✓ **Data quality checks:** Missing values, outliers, duplicates
- ✓ **Train test split early:** Stratified split for classification
- ✓ **Avoid data leakage:** Isolate test set completely during processing

Feature Optimization

Feature Engineering



Create and transform features to expose patterns to the model.

Feature Scaling



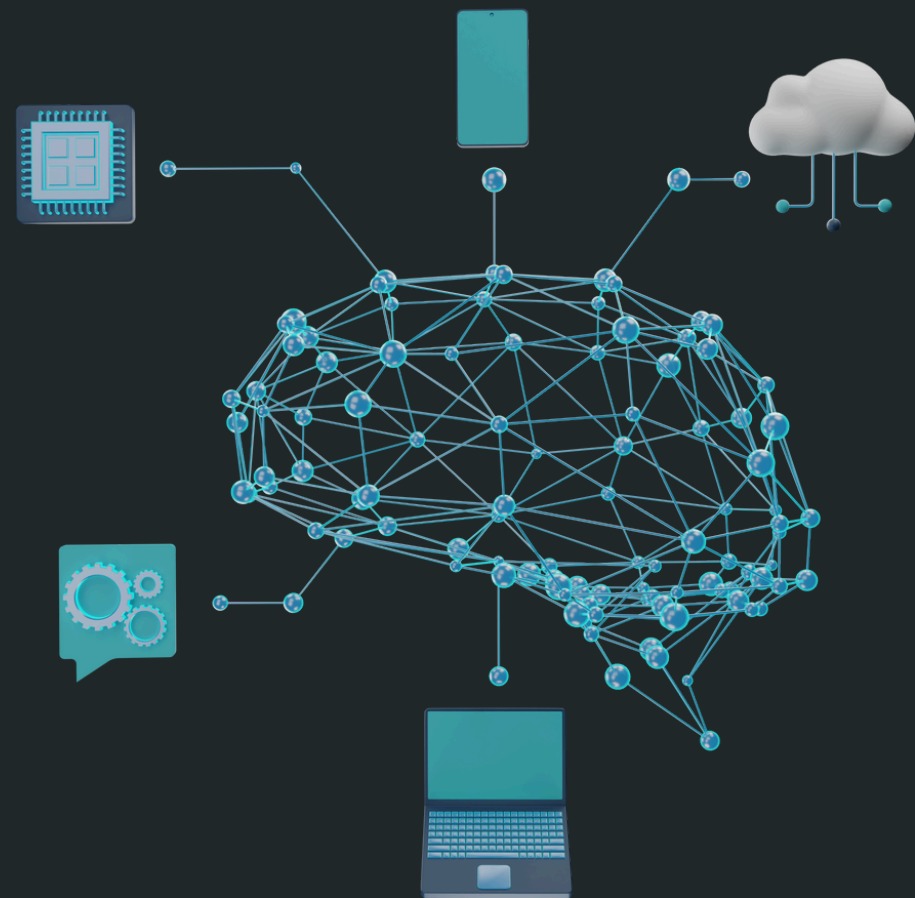
Scale using training data only to maintain distribution consistency.

Feature Selection



Remove irrelevant or redundant features to reduce noise.

Model Optimization



Model Optimization

- ✓ **Handle class imbalance** Balance class distribution
- ✓ **Model selection** Compare multiple algorithms
- ✓ **Hyperparameter tuning** Grid or random search
- ✓ **Regularization** Control model complexity

Validation & Evaluation

Cross Validation



Use cross validation for a reliable performance estimate.

Evaluation Metric



Choose correctly: Accuracy vs Precision, Recall, F1, or ROC AUC.

Error Analysis



Deeply analyze wrong predictions to understand model weaknesses.